

ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

— * —



BÁO CÁO BÀI TẬP 19/5

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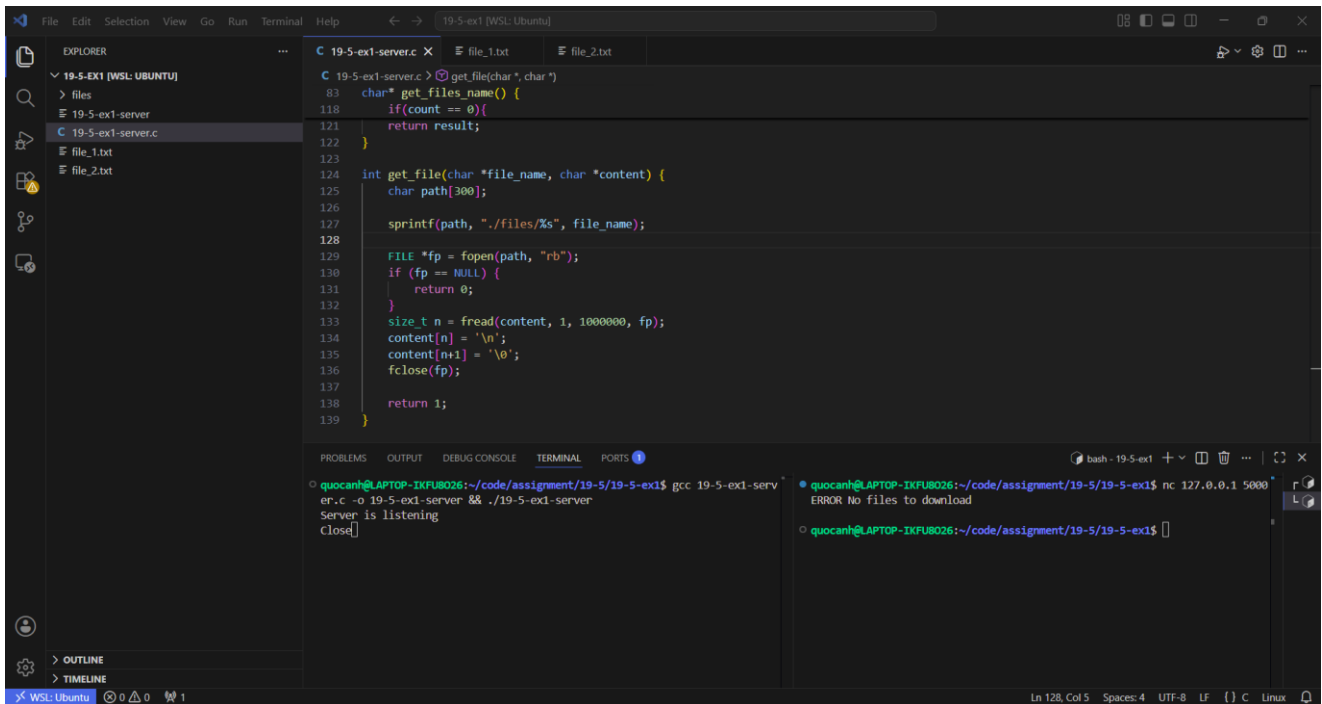
Học Phần: IT4060

Mã Lớp: 168503

Github: <https://github.com/quocanh13/IT4060>

Bài 1: Sử dụng kỹ thuật đa tiến trình, lập trình ứng dụng file Server

- Khi danh sách file trong folder files rỗng (các file nằm ngoài folder) thì server báo lỗi và đóng kết nối

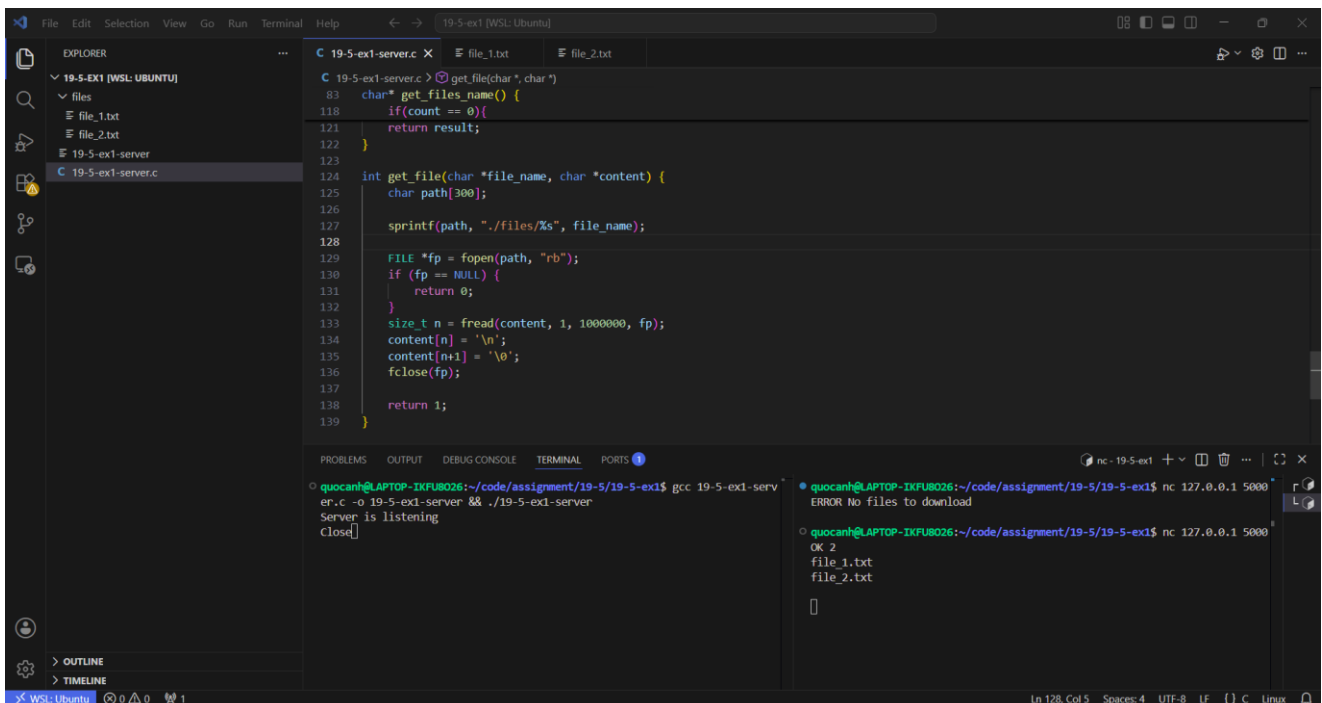


```
19-5-ex1-server.c
83 char* get_files_name() {
118 if(count == 0){
121 return result;
122 }
123
124 int get_file(char *file_name, char *content) {
125 char path[300];
126
127 sprintf(path, "./files/%s", file_name);
128
129 FILE *fp = fopen(path, "rb");
130 if (fp == NULL) {
131 return 0;
132 }
133 size_t n = fread(content, 1, 1000000, fp);
134 content[n] = '\n';
135 content[n+1] = '\0';
136 fclose(fp);
137
138 return 1;
139 }
```

```
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex1$ gcc 19-5-ex1-server.c -o 19-5-ex1-server && ./19-5-ex1-server
Server is listening
Close
```

```
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex1$ nc 127.0.0.1 5000
ERROR No files to download
```

- Khi di chuyển các file vào folder files thì server phản hồi hợp lệ

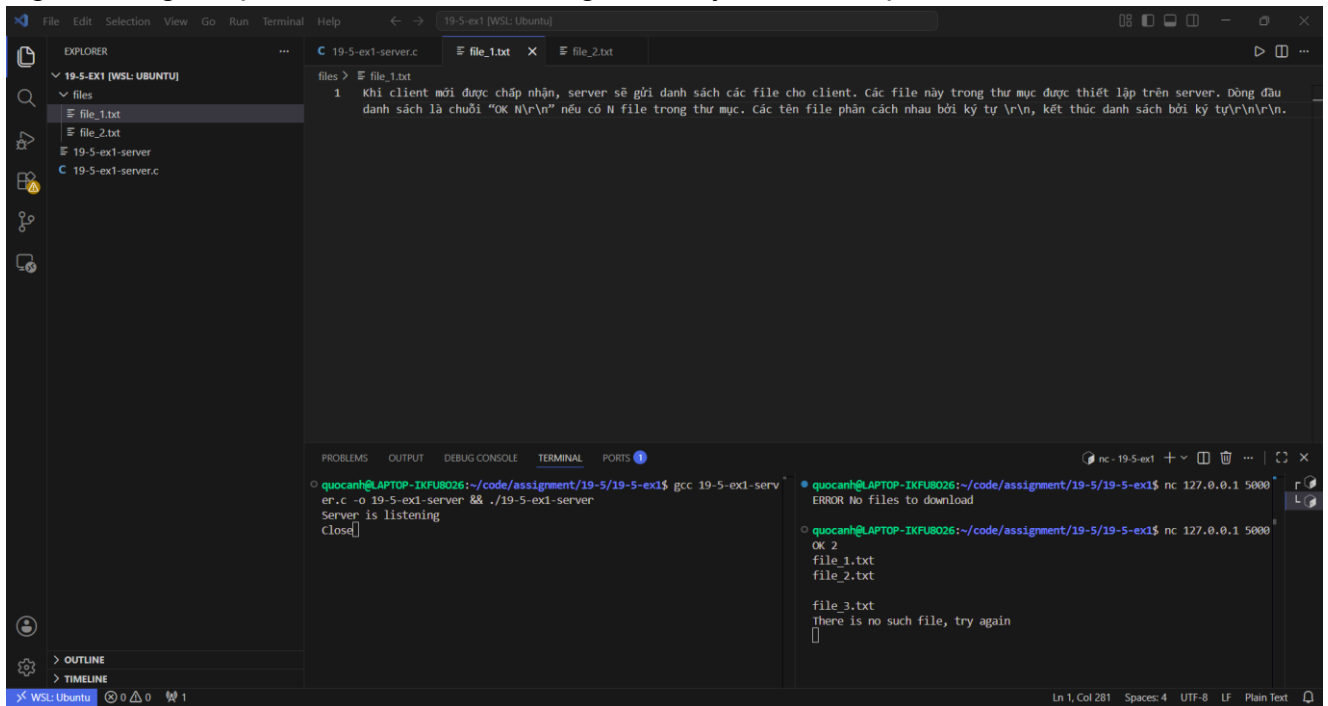


```
19-5-ex1-server.c
83 char* get_files_name() {
118 if(count == 0){
121 return result;
122 }
123
124 int get_file(char *file_name, char *content) {
125 char path[300];
126
127 sprintf(path, "./files/%s", file_name);
128
129 FILE *fp = fopen(path, "rb");
130 if (fp == NULL) {
131 return 0;
132 }
133 size_t n = fread(content, 1, 1000000, fp);
134 content[n] = '\n';
135 content[n+1] = '\0';
136 fclose(fp);
137
138 return 1;
139 }
```

```
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex1$ gcc 19-5-ex1-server.c -o 19-5-ex1-server && ./19-5-ex1-server
Server is listening
Close
```

```
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex1$ nc 127.0.0.1 5000
OK 2
file_1.txt
file_2.txt
```

- Người dùng nhập sai tên file server thông báo và yêu cầu nhập lại



```
1 Khi client mới được chấp nhận, server sẽ gửi danh sách các file cho client. Các file này trong thư mục được thiết lập trên server. Dòng đầu
danh sách là chuỗi "OK N\r\n" nếu có N file trong thư mục. Các tên file phân cách nhau bởi ký tự \r\n, kết thúc danh sách bởi ký tự\r\n\r\n.
```

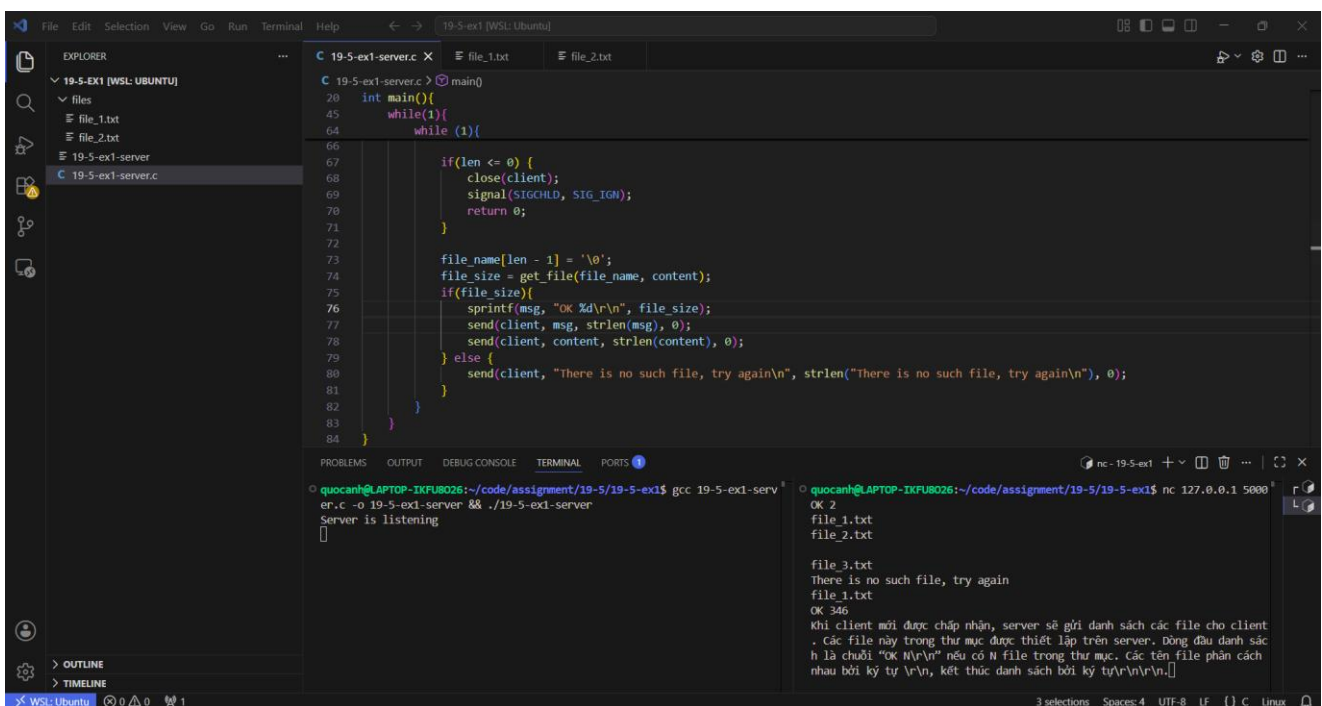
```
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex1$ gcc 19-5-ex1-server.c -o 19-5-ex1-server && ./19-5-ex1-server
Server is listening
Close]

quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex1$ nc 127.0.0.1 5000
ERROR: No files to download

quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex1$ nc 127.0.0.1 5000
OK 2
file 1.txt
file 2.txt

file 3.txt
There is no such file, try again
[]
```

- Người dùng nhập đúng tên file và server OK N\r\n và nội dung file cho người dùng



```
20 int main(){
45     while(1){
64         while(1){
66             if(len <= 0) {
67                 close(client);
68                 signal(SIGCHLD, SIG_IGN);
69                 return 0;
70             }
71         }
72
73         file_name[len - 1] = '\0';
74         file_size = get_file(file_name, content);
75         if(file_size){
76             sprintf(msg, "OK %d\r\n", file_size);
77             send(client, msg, strlen(msg), 0);
78             send(client, content, strlen(content), 0);
79         } else {
80             send(client, "There is no such file, try again\r\n", strlen("There is no such file, try again\r\n"), 0);
81         }
82     }
83 }
84 }
```

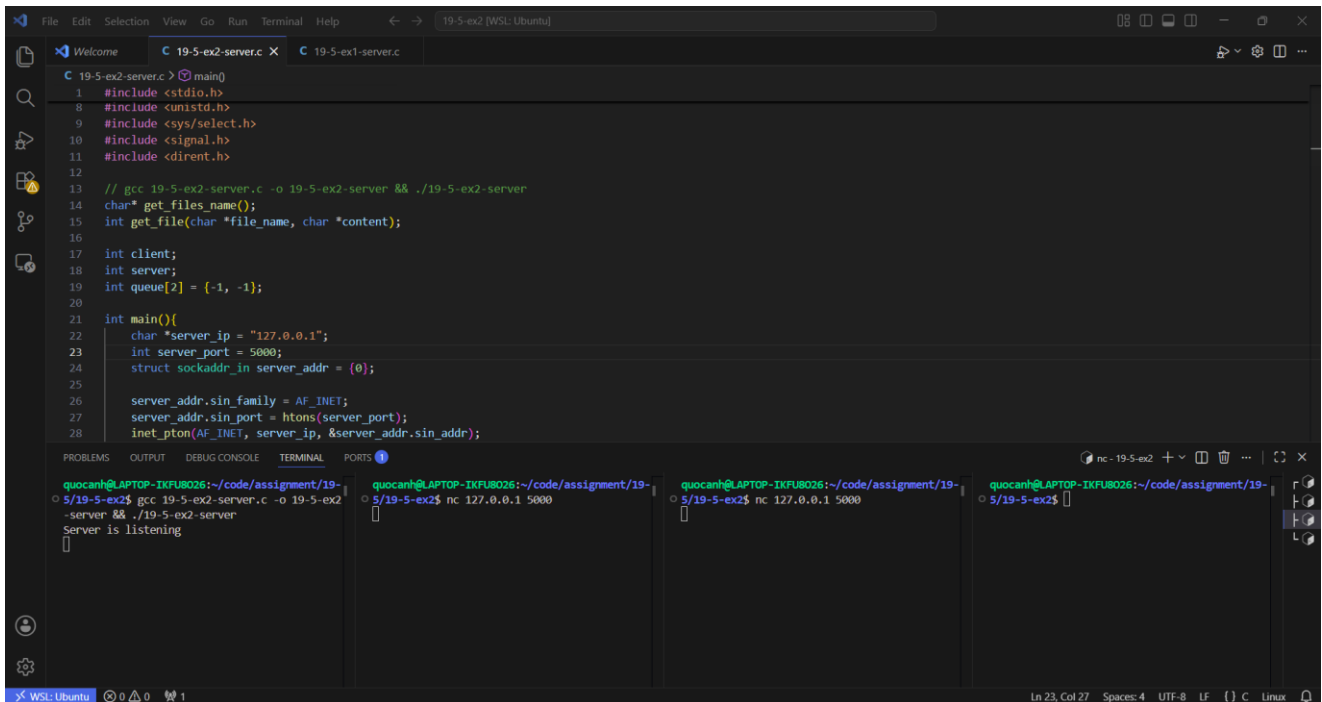
```
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex1$ gcc 19-5-ex1-server.c -o 19-5-ex1-server && ./19-5-ex1-server
Server is listening

quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex1$ nc 127.0.0.1 5000
OK 2
file 1.txt
file 2.txt

file 3.txt
There is no such file, try again
file 1.txt
OK 346
Khi client mới được chấp nhận, server sẽ gửi danh sách các file cho client
. Các file này trong thư mục được thiết lập trên server. Dòng đầu danh sách
h là chuỗi "OK N\r\n" nếu có N file trong thư mục. Các tên file phân cách
nhau bởi ký tự \r\n, kết thúc danh sách bởi ký tự\r\n\r\n.
```

Bài 2: Sử dụng kỹ thuật đa luồng, lập trình ứng dụng chat nhóm 2 người

- Client 1 và client 2 kết nối tới server tạo thành 1 cặp



```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <sys/select.h>
4 #include <signal.h>
5 #include <dirent.h>
6
7 // gcc 19-5-ex2-server.c -o 19-5-ex2-server && ./19-5-ex2-server
8 char* get_files_name();
9 int get_file(char *file_name, char *content);
10
11 int client;
12 int server;
13 int queue[2] = {-1, -1};
14
15 int main()
16 {
17     char *server_ip = "127.0.0.1";
18     int server_port = 5000;
19     struct sockaddr_in server_addr = {0};
20
21     server_addr.sin_family = AF_INET;
22     server_addr.sin_port = htons(server_port);
23     inet_pton(AF_INET, server_ip, &server_addr.sin_addr);
24 }
```

quocanh@LAPTOP-IXFU8026:~/code/assignment/19-5-ex2\$ gcc 19-5-ex2-server.c -o 19-5-ex2-server && ./19-5-ex2-server

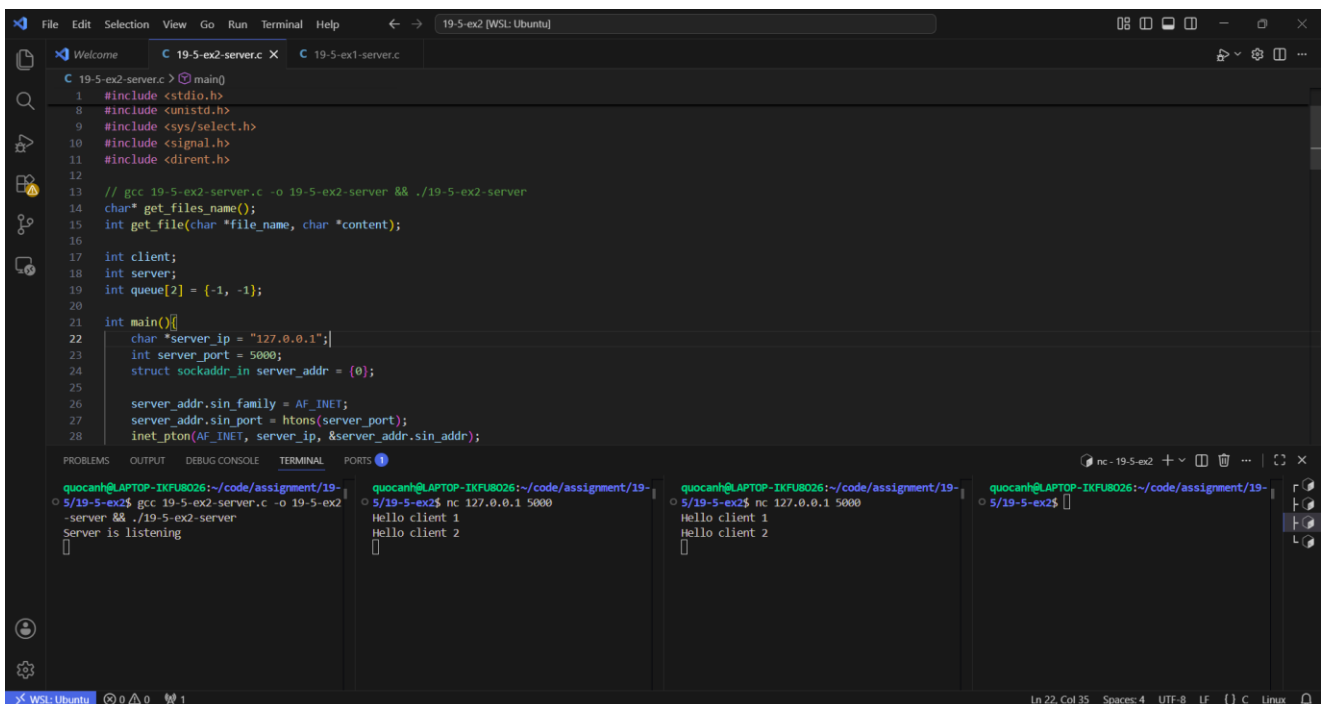
Server is listening

quocanh@LAPTOP-IXFU8026:~/code/assignment/19-5-ex2\$ nc 127.0.0.1 5000

quocanh@LAPTOP-IXFU8026:~/code/assignment/19-5-ex2\$ nc 127.0.0.1 5000

quocanh@LAPTOP-IXFU8026:~/code/assignment/19-5-ex2\$

- Client 1 và 2 gửi tin nhắn qua lại cho nhau



```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <sys/select.h>
4 #include <signal.h>
5 #include <dirent.h>
6
7 // gcc 19-5-ex2-server.c -o 19-5-ex2-server && ./19-5-ex2-server
8 char* get_files_name();
9 int get_file(char *file_name, char *content);
10
11 int client;
12 int server;
13 int queue[2] = {-1, -1};
14
15 int main()
16 {
17     char *server_ip = "127.0.0.1";
18     int server_port = 5000;
19     struct sockaddr_in server_addr = {0};
20
21     server_addr.sin_family = AF_INET;
22     server_addr.sin_port = htons(server_port);
23     inet_pton(AF_INET, server_ip, &server_addr.sin_addr);
24 }
```

quocanh@LAPTOP-IXFU8026:~/code/assignment/19-5-ex2\$ gcc 19-5-ex2-server.c -o 19-5-ex2-server && ./19-5-ex2-server

Server is listening

quocanh@LAPTOP-IXFU8026:~/code/assignment/19-5-ex2\$ nc 127.0.0.1 5000

Hello client 1

quocanh@LAPTOP-IXFU8026:~/code/assignment/19-5-ex2\$ nc 127.0.0.1 5000

Hello client 1

quocanh@LAPTOP-IXFU8026:~/code/assignment/19-5-ex2\$

- Client 3 kết nối tới server

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <sys/select.h>
4 #include <signal.h>
5 #include <dirent.h>
6
7 // gcc 19-5-ex2-server.c -o 19-5-ex2-server && ./19-5-ex2-server
8 char* get_files_name();
9 int get_file(char *file_name, char *content);
10
11 int client;
12 int server;
13 int queue[2] = {-1, -1};
14
15 int main(){
16     char *server_ip = "127.0.0.1";
17     int server_port = 5000;
18     struct sockaddr_in server_addr = {0};
19
20     server_addr.sin_family = AF_INET;
21     server_addr.sin_port = htons(server_port);
22     inet_pton(AF_INET, server_ip, &server_addr.sin_addr);
23
24     quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5-ex2$ gcc 19-5-ex2-server.c -o 19-5-ex2-server && ./19-5-ex2-server
25 Server is listening
26
27 quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5-ex2$ nc 127.0.0.1 5000
28 Hello client 1
29 Hello client 2
```

- Do client 1 và 2 nhắn với nhau tiếp client 3 không nhận được do không phải là cặp với 1 và 2

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <sys/select.h>
4 #include <signal.h>
5 #include <dirent.h>
6
7 // gcc 19-5-ex2-server.c -o 19-5-ex2-server && ./19-5-ex2-server
8 char* get_files_name();
9 int get_file(char *file_name, char *content);
10
11 int client;
12 int server;
13 int queue[2] = {-1, -1};
14
15 int main(){
16     char *server_ip = "127.0.0.1";
17     int server_port = 5000;
18     struct sockaddr_in server_addr = {0};
19
20     server_addr.sin_family = AF_INET;
21     server_addr.sin_port = htons(server_port);
22     inet_pton(AF_INET, server_ip, &server_addr.sin_addr);
23
24     quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5-ex2$ gcc 19-5-ex2-server.c -o 19-5-ex2-server && ./19-5-ex2-server
25 Server is listening
26
27 quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5-ex2$ nc 127.0.0.1 5000
28 Hello client 1
29 Hello client 2
30 Hello
```

- Thêm client 4 kết nối đến server tạo thành 1 cặp với client 3 và gửi nhận tin nhắn cho nhau. Client 1 và 2 không thấy tin nhắn đó.

The screenshot shows a VS Code editor with a C program named `19-5-ex2-server.c`. The program is a multi-threaded server that listens on port 5000 and handles incoming connections. It uses `pthread_create` to spawn threads for each client. The code includes headers for `stdio.h`, `unistd.h`, `sys/select.h`, `signal.h`, and `dirent.h`. It defines a `get_files_name` function and a `main` function that sets up the server and handles clients.

Below the editor, there are five terminal windows showing the execution of the program. The first terminal shows the server listening. The other four terminals show the server receiving connections from clients 1, 2, 3, and 4. Client 1 and 2 send "Hello" messages, while client 3 sends "Hello client 4" and client 4 sends "Hello".

Bài 3: Lập trình ứng dụng chat_server (slide 173) sử dụng kỹ thuật multithread.

- Client nhập sai format, server không chấp nhận và yêu cầu nhập lại

The screenshot shows a VS Code editor with a C program named `19-5-ex3-server.c`. The program is a multi-threaded server that listens on port 5000 and handles incoming connections. It uses `pthread_create` to spawn threads for each client. The code includes headers for `stdio.h`, `unistd.h`, `sys/select.h`, `signal.h`, and `dirent.h`. It defines a `get_files_name` function and a `main` function that sets up the server and handles clients.

Below the editor, there are three terminal windows showing the execution of the program. The first terminal shows the server listening. The other two terminals show the server receiving connections from clients 1 and 2. Client 1 sends "Invalid format" and "Try again with the following format: client_id: client_name". Client 2 sends "Invalid format" and "Try again with the following format: client_id: client_name".

- Client nhập đúng format của id, server chấp nhận và không báo lỗi, có thể bắt đầu nhắn tin với nhau

The screenshot shows a VS Code editor with a C file named `19-5-ex3-server.c`. The code defines a `create_thread` function that sends a message to all clients and a `check_name` function that validates the client ID format. The code is as follows:

```
1 19-5-ex3-server.c > check_name(char [], int, int)
2 void* create_thread(void* args){
3     while(1){
4         if(strcmp(client_names[client], "") == 0){
5             msg[len] = '\0';
6             char final_message[2000];
7             sprintf(final_message, "%s: %s", client_names[client], msg);
8             for(int i = 0; i < 1000; i++){
9                 if(strcmp(client_names[i], "") && strcmp(client_names[i], client_names[client])){
10                     send(i, final_message, strlen(final_message), 0);
11                 }
12             }
13         }
14     }
15 }
16
17 int check_name(char name[], int len, int client){
18     name[len - 1] = '\0';
19     char *tok = strtok(name, " ");
20     if(tok == NULL || strcmp(tok, "client_id:")) {
21         char *message = "Invalid format\nTry again with the following format: client_id: client_name\n";
22         send(client, message, strlen(message), 0);
23         return 0;
24     }
25 }
```

Below the editor are three terminal windows:

- Terminal 1:** Shows the server starting and listening on port 5000.
- Terminal 2:** A client (nc 1) connects to 27.0.0.1 5000. It enters "client_id: client_1" and receives "Invalid format" and "Try again with the following format: client_id: client_name". It then enters "client_id:client_1" and receives "Invalid format" and "Try again with the following format: client_id: client_name". Finally, it enters "client_id: client_1" and receives "Hello".
- Terminal 3:** A client (nc 12) connects to 7.0.0.1 5000. It enters "client_id client_2" and receives "Invalid format" and "Try again with the following format: client_id: client_name". It then enters "client-id: client_2" and receives "Invalid format" and "Try again with the following format: client_id: client_name". Finally, it enters "client_id: client_2" and receives "Hello".

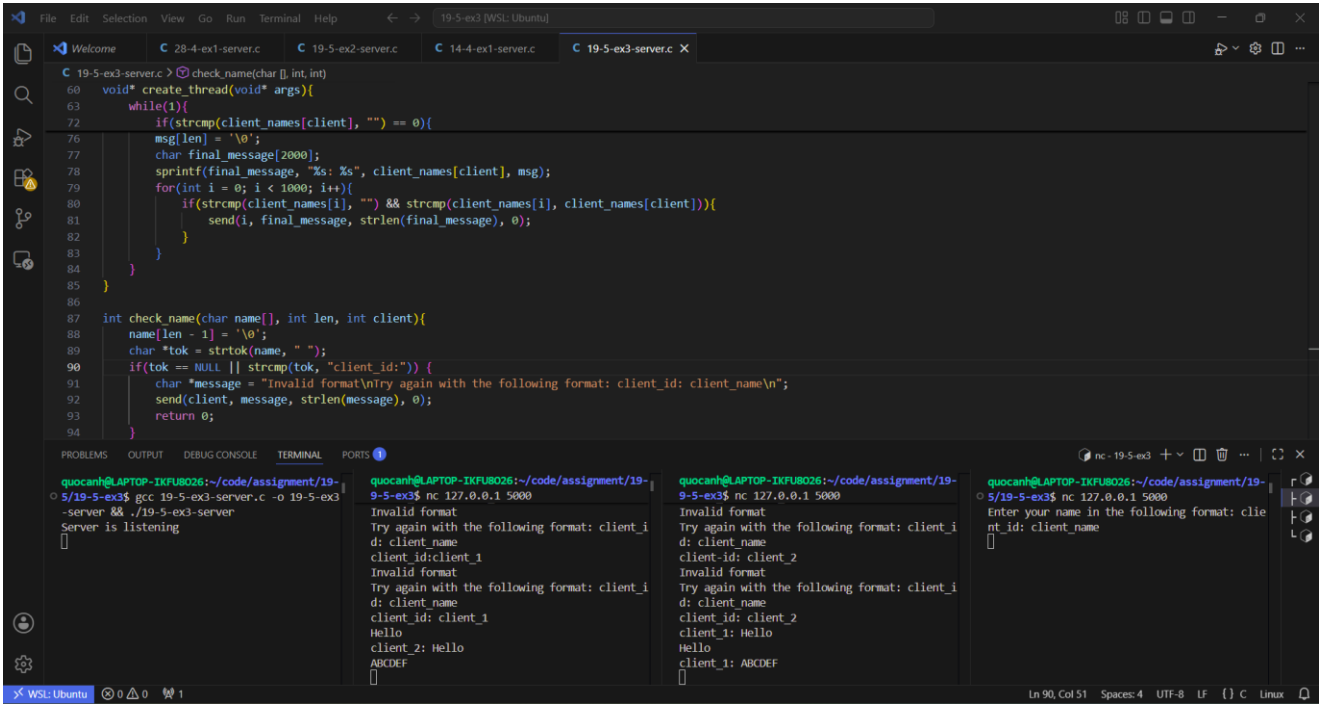
- Client nhắn tin với nhau

This screenshot shows the same setup as the previous one, but the clients are now sending messages to each other. The code in the editor remains the same.

The terminal windows show the following interactions:

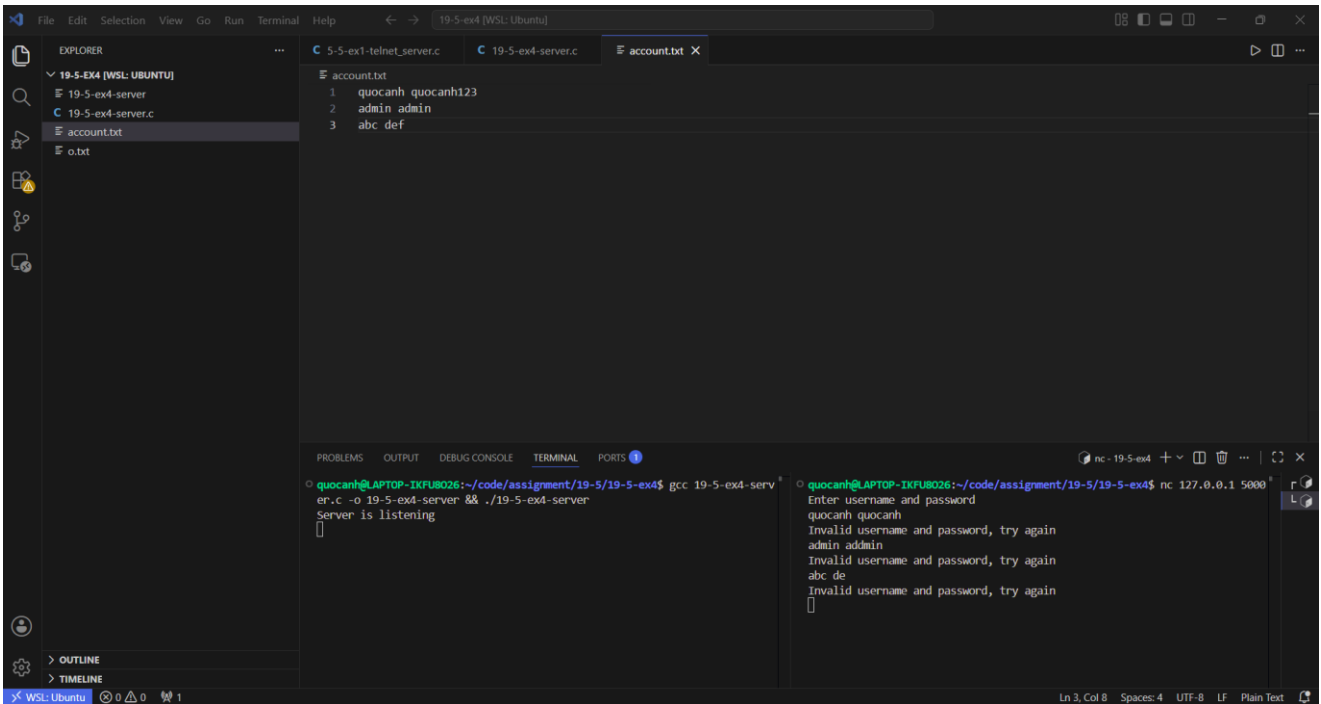
- Terminal 1:** The server is still listening.
- Terminal 2:** The client (nc 1) sends "Hello" to the server and receives "Hello". It then sends "client_2: Hello" to the server and receives "Hello".
- Terminal 3:** The client (nc 12) sends "Hello" to the server and receives "Hello". It then sends "client_1: Hello" to the server and receives "Hello".

- Client mới chưa nhập đúng tên không nhận được tin nhắn



Bài 4: Lập trình ứng dụng telnet_server (slide 174) sử dụng kỹ thuật multithread.

- Client nhập sai usernam và password server không chấp nhận vào yêu cầu nhập lại



- Client nhập đúng username và password được server chấp nhận và bắt đầu gửi command

The screenshot shows the VS Code interface with the following content:

EXPLORER:

- 19-5-EX4 [WSL: UBUNTU]
 - 19-5-ex4-server
 - 19-5-ex4-server.c
 - account.txt
 - o.txt

account.txt:

```
1 quocanh quocanh123
2 admin admin
3 abc def
```

TERMINAL:

```
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex4$ gcc 19-5-ex4-server.c -o 19-5-ex4-server && ./19-5-ex4-server
Server is listening

nc - 19-5-ex4$ nc 127.0.0.1 5000
Enter username and password
quocanh quocanh
Invalid username and password, try again
admin admin
Invalid username and password, try again
abc de
Invalid username and password, try again
quocanh quocanh123
Logged in successfully
```

- Client gửi command “ls -l” và “pwd” cho server, server thực thi và gửi lại kết quả cho client

The screenshot shows the VS Code interface with the following content:

EXPLORER:

- 19-5-EX4 [WSL: UBUNTU]
 - 19-5-ex4-server
 - 19-5-ex4-server.c
 - account.txt
 - o.txt

account.txt:

```
1 quocanh quocanh123
2 admin admin
3 abc def
```

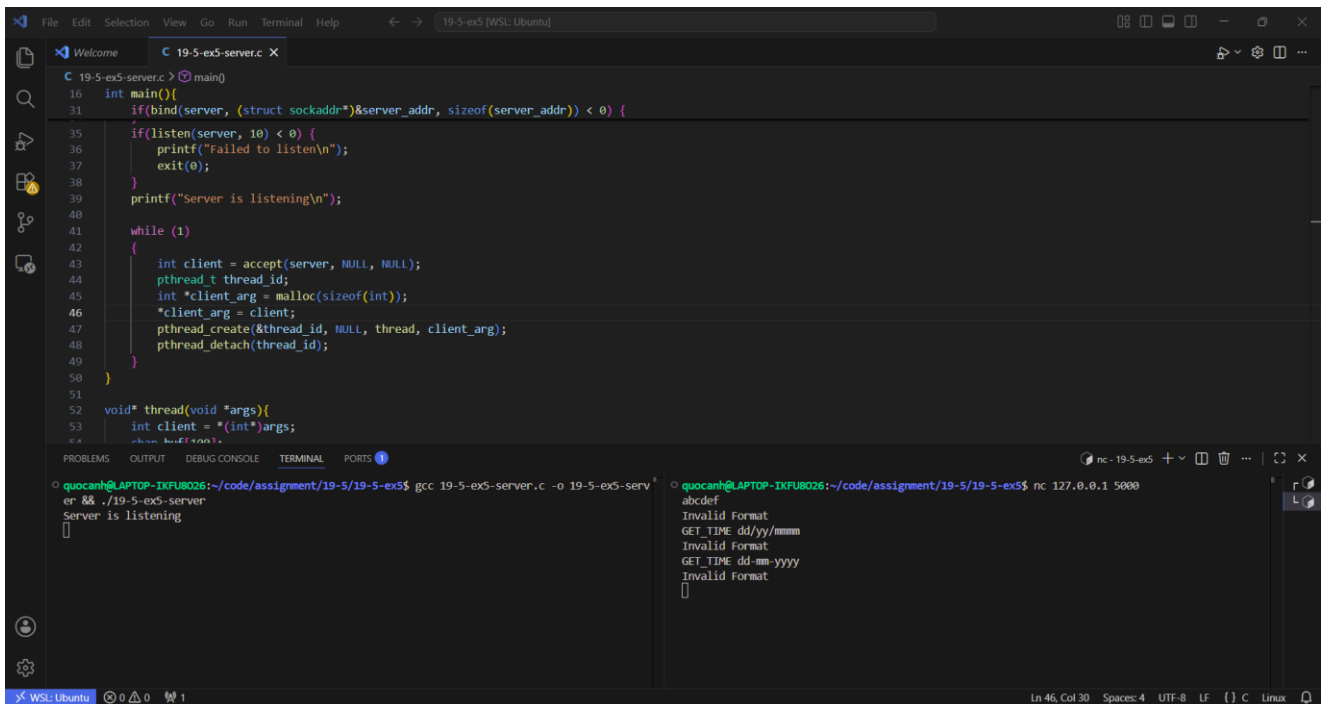
TERMINAL:

```
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex4$ gcc 19-5-ex4-server.c -o 19-5-ex4-server && ./19-5-ex4-server
Server is listening

nc - 19-5-ex4$ nc 127.0.0.1 5000
abc de
Invalid username and password, try again
quocanh quocanh123
Logged in successfully
ls -l
total 28
-rwxr-xr-x 1 quocanh quocanh 17120 May 25 09:12 19-5-ex4-server
-rw-r--r-- 1 quocanh quocanh 3482 May 25 09:09 19-5-ex4-server.c
-rw-r--r-- 1 quocanh quocanh 38 May 25 09:10 account.txt
-rw-r--r-- 1 quocanh quocanh 0 May 25 09:13 o.txt
pwd
/home/quocanh/code/assignment/19-5/19-5-ex4
```

Bài 5: Lập trình ứng dụng time_server (slide 204) sử dụng kỹ thuật multithread.

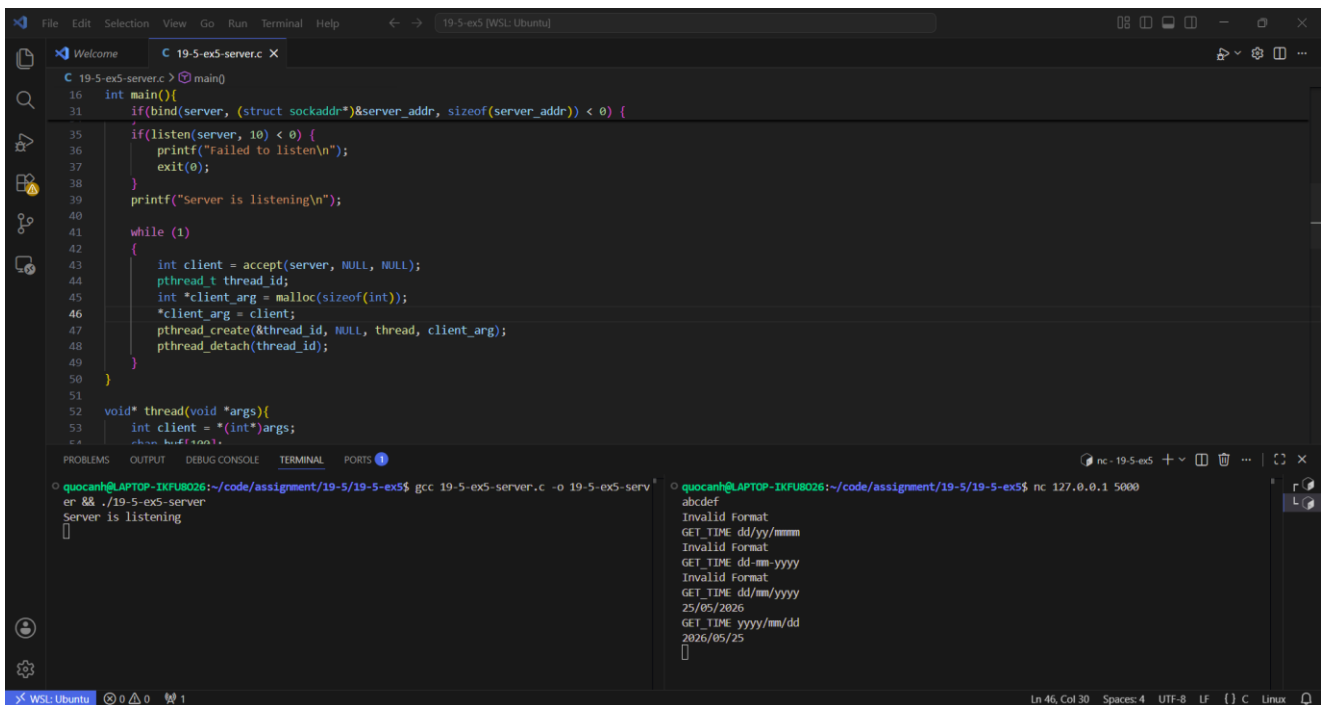
- Client gửi sai format server không chấp nhận



```
16 int main(){
31     if(bind(server, (struct sockaddr*)&server_addr, sizeof(server_addr)) < 0) {
35         if(listen(server, 10) < 0) {
36             printf("Failed to listen\n");
37             exit(0);
38         }
39         printf("Server is listening\n");
40     }
41     while (1)
42     {
43         int client = accept(server, NULL, NULL);
44         pthread_t thread_id;
45         int *client_arg = malloc(sizeof(int));
46         *client_arg = client;
47         pthread_create(&thread_id, NULL, thread, client_arg);
48         pthread_detach(thread_id);
49     }
50 }
51
52 void* thread(void *args){
53     int client = *(int*)args;
54     char buffer[1024];
```

```
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex5$ gcc 19-5-ex5-server.c -o 19-5-ex5-server
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex5$ nc 127.0.0.1 5000
abcdef
Invalid Format
GET_TIME dd/yy/mm
Invalid Format
GET_TIME dd-mm-yyyy
Invalid Format
```

- Client gửi đúng format, server chấp nhận và phải trả kết quả tương ứng

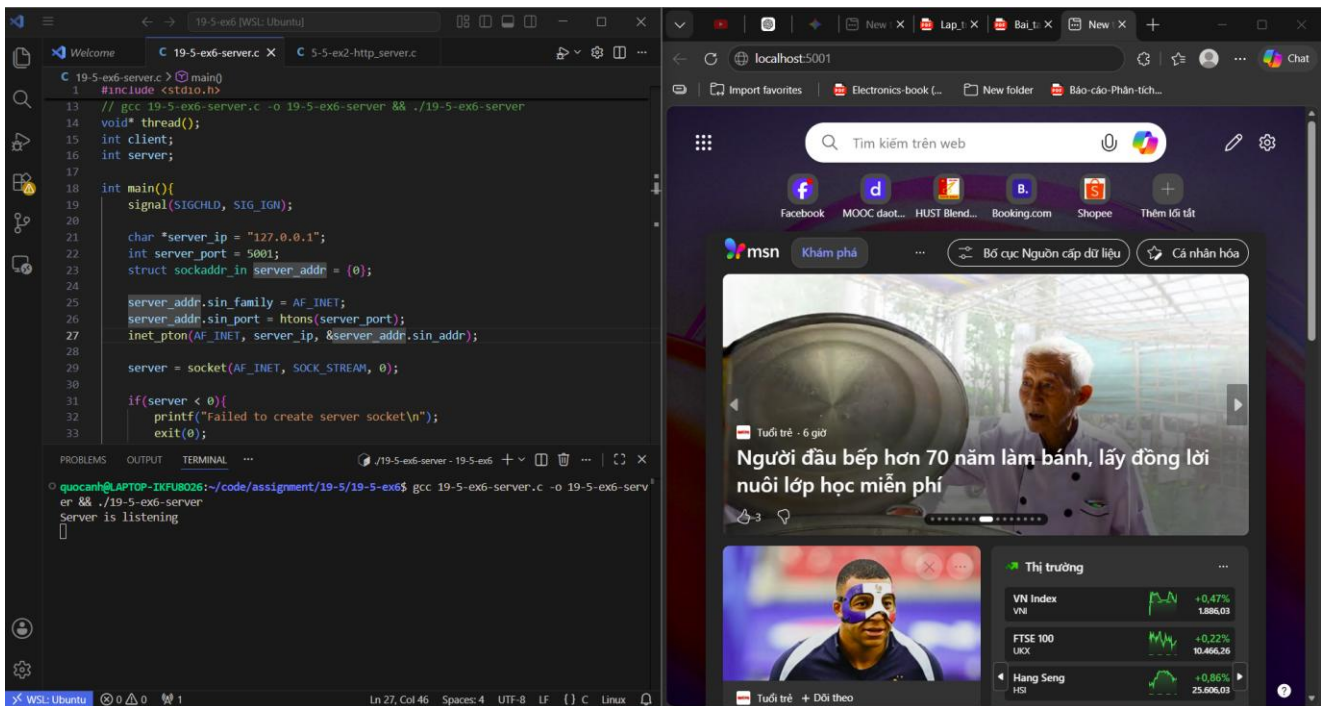


```
16 int main(){
31     if(bind(server, (struct sockaddr*)&server_addr, sizeof(server_addr)) < 0) {
35         if(listen(server, 10) < 0) {
36             printf("Failed to listen\n");
37             exit(0);
38         }
39         printf("Server is listening\n");
40     }
41     while (1)
42     {
43         int client = accept(server, NULL, NULL);
44         pthread_t thread_id;
45         int *client_arg = malloc(sizeof(int));
46         *client_arg = client;
47         pthread_create(&thread_id, NULL, thread, client_arg);
48         pthread_detach(thread_id);
49     }
50 }
51
52 void* thread(void *args){
53     int client = *(int*)args;
54     char buffer[1024];
```

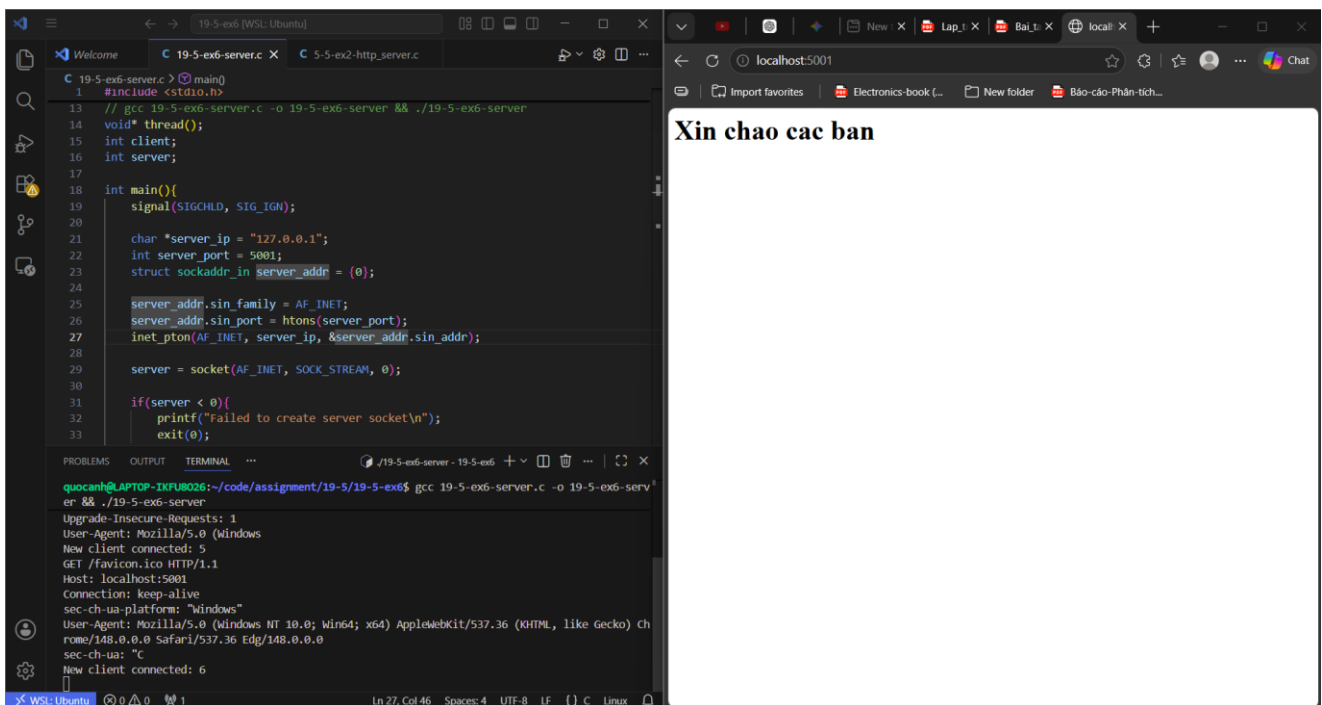
```
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex5$ gcc 19-5-ex5-server.c -o 19-5-ex5-server
quocanh@LAPTOP-IXFUB026:~/code/assignment/19-5/19-5-ex5$ nc 127.0.0.1 5000
abcdef
Invalid Format
GET_TIME dd/yy/mm
Invalid Format
GET_TIME dd-mm-yyyy
Invalid Format
GET_TIME dd/mm/yyyy
25/05/2026
GET_TIME yyyy/mm/dd
2026/05/25
```

Bài 6: Lập trình ứng dụng http_server (slide 154) sử dụng kỹ thuật prethreading.

- Chạy server



- Truy cập vào server từ web browser, server nhận được request và phản hồi



- Thêm một kết nối nữa

